

Customer Shopping Behavior
CCDS 321 – Data Mining
Milestone 4

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Dataset Description

This dataset describes customer shopping behavior, with 5,050 observations and 17 attributes, which contains 13 categorical and 4 numerical, covering customer information in detail such as age, gender, and location. In addition, it contains purchase details such as category, item purchased, and purchase amount, as well as behavioral attributes such as subscription status, purchase frequency, and payment method.

Data Preprocessing

- Missing Data Handling

The dataset had 5,050 records. Missing values in numerical variables (Purchase Amount, Review Rating, Previous Purchases) were replaced with the mean. Missing values in the categorical variable (Size) were replaced with the mode ("M"). After imputation, all missing values were filled, no rows were deleted, and original values remained unchanged.

- Duplicate Records Handling

The dataset was checked for fully duplicate records. A total of 50 duplicate rows were found and removed. After removing duplicates, the number of rows changed from 5,050 to 5,000, while the original non-duplicate records remained unchanged.

- Inconsistent Values

The categorical variables were checked for repeated labels with the same meaning. In the Season variable, Autumn was replaced with Fall. In the Frequency of Purchases variable, Fortnightly was replaced with Bi-Weekly, and Every 3 Months was replaced with Quarterly. This made the categories more consistent before modelling.

- Outlier Detection and Removal

Outliers were checked using the Purchase Amount variable. High purchase amount values were mainly found in shirts, Shoes, and watches. These outliers were removed using filter rows. After removing 132 outliers, the final dataset contained 4,868 rows.

Task 1

Percentages: Training: 70, Test: 30, Validation: 30, Method: Stratified

Splitting Criterion	CHAID
	Entropy
	Gini
	Information gain ratio
Minimum Number of Observations per Leaf (Leaf Size)	3.5% = 119
	6% = 204
Maximum number of branches:	2
	4

1. Evaluation Approach

Measure	Description	Definition of Value Function	Weight	Threshold
Accuracy	Overall proportion of correct predictions made by the model on the test partition	Accuracy Score = Accuracy	0.60	0.70
Simplicity	Measures tree complexity based on the size of the tree (number of leaves)	Simplicity Score = $1 - (\text{Leaves} - \text{Min Leaves}) / (\text{Max Leaves} - \text{Min Leaves})$	0.05	-
Lift	Measures the model's power at the 2nd decile compared to a random guess	Lift Score = $(\text{Lift Value} - \text{Min Lift}) / (\text{Max Lift} - \text{Min Lift})$	0.20	-
Stability	Measures the consistency of the DT performance between Training and Test sets for target event Subscription status	Stability Score = 1 if $\text{MR_Training} - \text{MR_Testing} \leq 0.05$ 0, Otherwise	0.15	1

Composite Performance Score Formula:

Overall Score

2. Summary of Results

DT Label/ Description	Performance Measures								Overall Score
	Accuracy		Simplicity		Lift		Stability		
	% Error	Score	No of Leaves	Score	Value	Score	Score	Justification	
DT CHAID 3.5% 2	23.15%	0.7685	2	1	1.8918	0	1	MR difference is 0.0048, below threshold of 0.05. Model is stable.	0.6611
DT1 CHAID 3.5% 4	23.15%	0.7685	5	0.6	1.8918	0	1	MR difference is 0.0048, below threshold of 0.05. Model is stable.	0.6411
DT2 CHAID 6% 2	23.15%	0.7685	2	1	1.8918	0	1	MR difference is 0.0048, below threshold of 0.05. Model is stable.	0.6611
DT3 CHAID 6% 4	23.15%	0.7685	2	1	1.8918	0	1	MR difference is 0.0048, below threshold of 0.05. Model is stable.	0.6611
DT4 Entropy 3.5% 2	23.68%	0.7632	6	0.3	1.9743	0.54	1	MR difference is 0.0238, below threshold of 0.05. Model is stable.	0.730
DT5 Entropy 3.5% 4	23.68%	0.7632	10	0	2.0135	0.54	1	MR difference is 0.0238, below threshold of 0.05. Model is stable.	0.716
DT6 Entropy 6% 2	23.15%	0.7685	5	0.6	1.9743	0.54	1	MR difference is 0.0090, below threshold of 0.05. Model is stable.	0.749
DT7 Entropy 6% 4	23.15%	0.7685	5	0.6	2.0444	1	1	MR difference is 0.0090, below threshold of 0.05. Model is stable.	0.841
DT8 GINI 3.5% 2	23.68%	0.7632	7	0.3	1.9743	0.540	1	MR difference is 0.0238, below threshold of 0.05. Model is stable.	0.730

DT9 GINI 3.5% 4	23.68%	0.7632	10	0	2.0135	0.797	1	MR difference is 0.0238, below threshold of 1. Model is stable.	0.767
DT10 GINI 6% 2	23.15%	0.7685	5	0.6	1.9743	0.540	1	MR difference is 0.0090, below threshold of 0.05. Model is stable.	0.749
DT11 GINI 6% 4	23.15%	0.7685	5	0.6	2.0444	1	1	MR difference is 0.0090, below threshold of 0.05. Model is stable.	0.841
DT12 INFO GAIN 3.5% 2	23.68%	0.7632	7	0.3	1.9743	0.540	1	MR difference is 0.0238, below threshold of 0.05. Model is stable.	0.730
DT13 INFO GAIN 3.5% 4	23.68%	0.7632	7	0.3	1.9743	0.540	1	MR difference is 0.0238, below threshold of 0.05. Model is stable.	0.730
DT14 INFO GAIN 6% 2	23.15%	0.7685	5	0.6	1.9743	0.540	1	MR difference is 0.0090, below threshold of 0.05. Model is stable.	0.749
DT15 INFO GAIN 6% 4	23.15%	0.7685	5	0.6	1.9743	0.540	1	MR difference is 0.0090, below threshold of 0.05. Model is stable.	0.749

The decision trees were evaluated using performance measures such as Accuracy, Simplicity, Lift, and Stability. The Lift values show that Entropy and Gini have the highest Lift value and CHAID has a lower value. However, the DT7 and DT11 get the best overall score of 0.841.

3. Evidence of Experimentation



Figure 1 - Process Flow Diagram

Data Pipelines Pipeline Comparison Insights								
Filter								
☒	Variable Name	Label	Type	Role	Assess for Bias	Level	Order	Comment
☐	Customer ID		Numeric	ID		Interval	Default	
☐	Discount Applied		Character	Input		Binary	Default	
☐	Frequency of Purchases		Character	Input		Nominal	Default	
☐	Gender		Character	Input		Nominal	Default	
☐	Item Purchased		Character	Rejected		Nominal	Default	The variable exc maximum numb cutoff value.
☐	Location		Character	Rejected		Nominal	Default	The variable exc maximum numb cutoff value.
☐	Payment Method		Character	Input		Nominal	Default	
☐	Previous Purchases		Numeric	Input		Interval	Default	
☐	Purchase Amount (USD)		Numeric	Input		Interval	Default	
☐	Review Rating		Numeric	Input		Interval	Default	
☐	Season		Character	Input		Nominal	Default	
☐	Shipping Type		Character	Input		Nominal	Default	
☐	Size		Character	Input		Nominal	Default	
☒	Subscription Status		Character	Target		Binary	Default	

Figure 2 - Variables Tab

Decision Tree (7) 🔍 ▶ 🖨️ ?

▼ Splitting Options

▼ Grow Criterion

Class target criterion:

Interval target criterion:

Significance level:

☐ Bonferroni

Maximum number of branches:
 2 4 6 10

Maximum depth:

Minimum leaf size:

Figure 3 - Tree Node Parameters

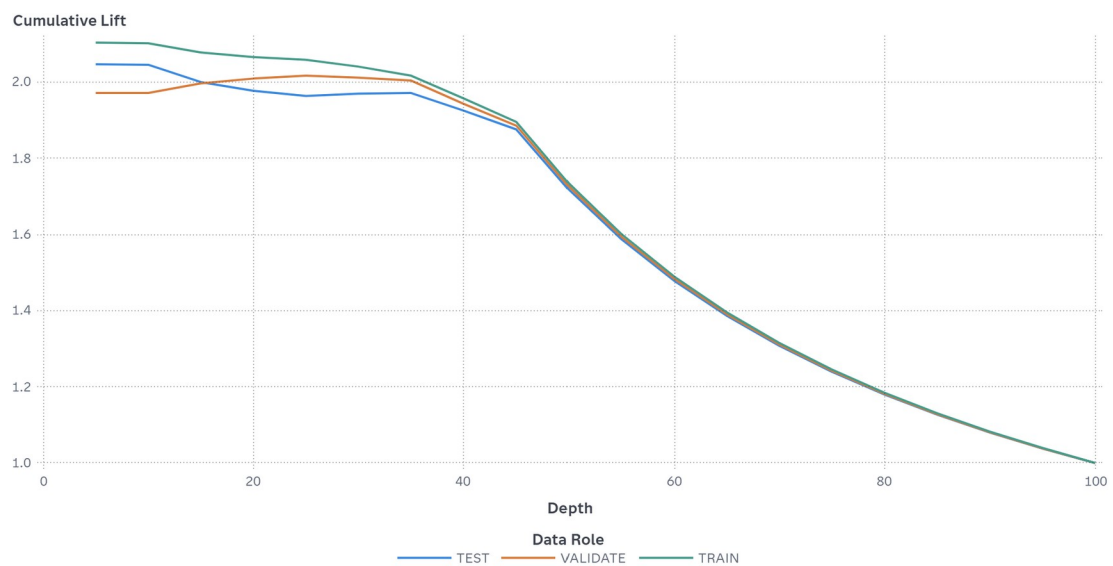


Figure 4 - Non-Cumulative %Lift Chart of Decision Tree 7

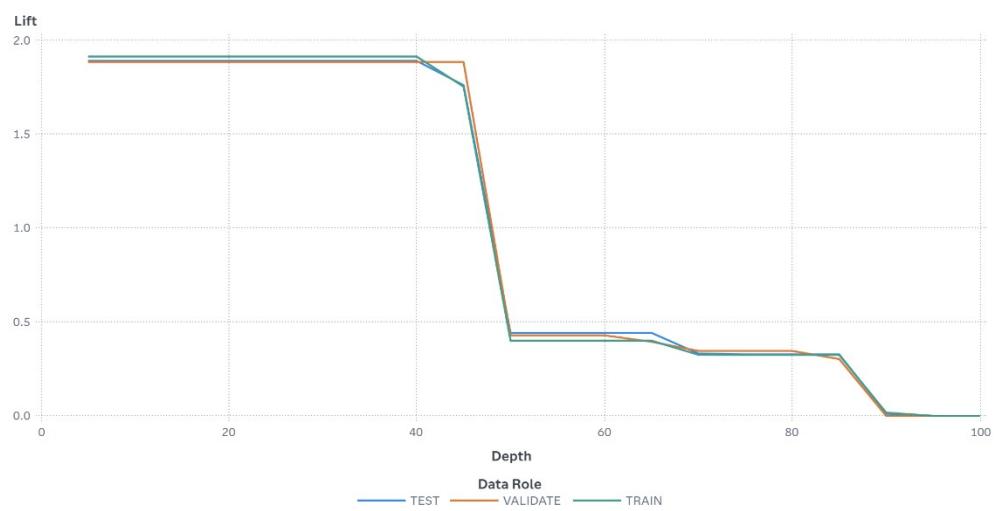
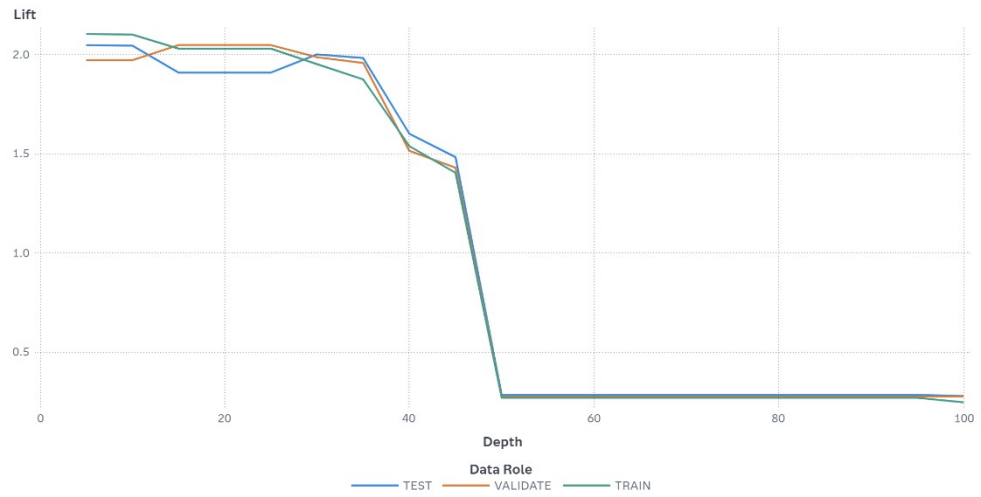


Figure 5 – Lift of Decision tree 7

Figure 6 - Lift of Decision Tree 1

The lift chart above compares DT1 and DT7 for the target event **Subscription Statues**. At the 2nd decile (depth of 20%), in DT1 the TRAIN partition has a Lift of 1.91, meaning there are 1.91 times more subscribers than random selection. Meanwhile, DT7 The TRAIN partition has a Lift of 2.1 at the same percentile, meaning there are 2.1 times more subscribers captured. In addition, DT7 shows consistent and stable Lift performance compared to DT1, which DT7 is a more stable model than DT1.

Tree Diagram

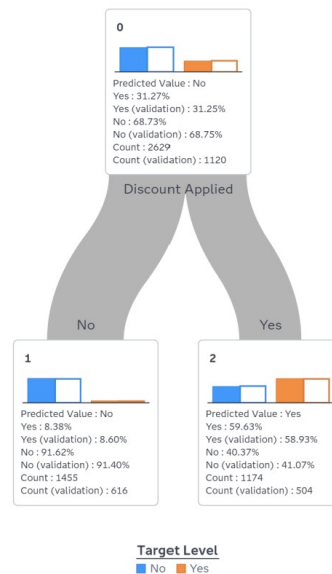


Figure 7 - Decision Tree Diagram 7

Different Combination

To find a better combination, a custom Decision Tree was built using:

- o Splitting Criterion: GINI
- o Maximum Number of Branches: 2
- o Maximum Tree Depth: 4
- o Minimum Leaf Size: 100

After calculating all metrics according to the defined value functions, we obtained the following results:

- o Accuracy Score = 0.7694
- o Simplicity Score = 0.9688
- o Lift Score = 1.494
- o Stability Score = 1

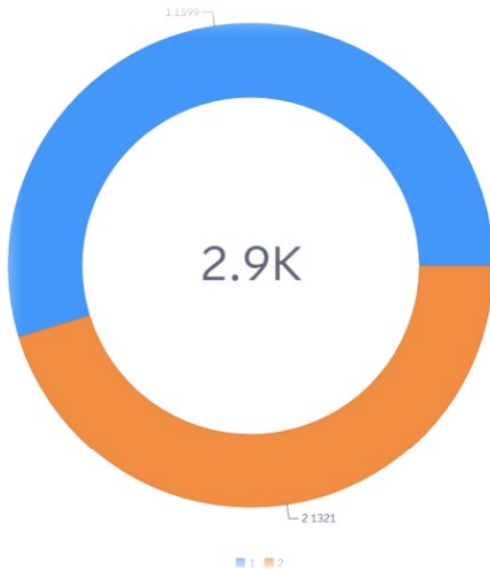
$$\text{Overall Score} = (0.60 \times 0.7694) + (0.05 \times 0.9688) + (0.20 \times 1.0) + (0.15 \times 1.0) = 0.860$$

Therefore, after applying this new combination, we achieved a better overall score (0.860) compared to the previous best model (0.841), making this the best performing model overall.

Task 2

1. Generate 4 Segmentations

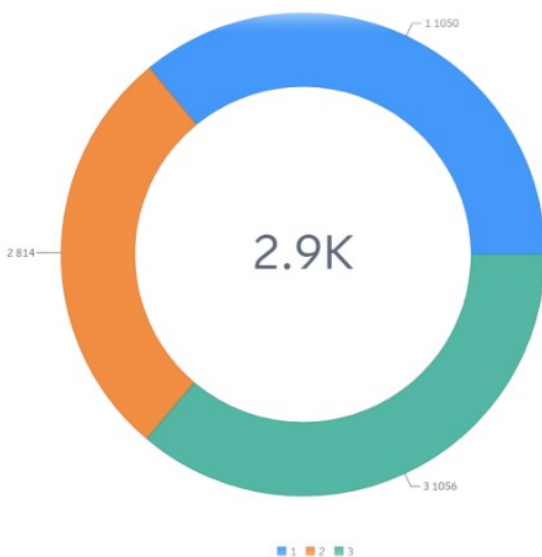
km_2 Segmentation:



Cluster	Size
Cluster 1	1599
Cluster 2	1321

The km_2 segmentation divided the customers into two clusters. **Cluster 1** contains 1599 customers, **Cluster 2** contains 1321 customers. The clusters are relatively balanced in size.

km_3 Segmentation:



Cluster	Size
Cluster 1	1050
Cluster 2	814
Cluster 3	1056

Cluster Centroids

Cluster ID	Age	Previous Purchases	Purchase Amount (USD)
1	47.0615	26.3086	101.1439
2	41.7440	21.9602	103.2497
3	43.0417	26.5538	169.0896

Review Rating	Standard: Age	Standard: Previous Purchases	Standard: Purchase Amount (USD)
3.5809	0.1964	0.0839	-0.1124
3.8150	-0.1515	-0.2337	-0.1032
3.6709	-0.0666	0.1018	0.1844

Standard: Review Rating	Category	Discount Applied	Frequency of Purchases
-0.1231	Clothing	No	Quarterly
0.1651	Accessories	No	Quarterly
-0.0123	Clothing	Yes	Quarterly

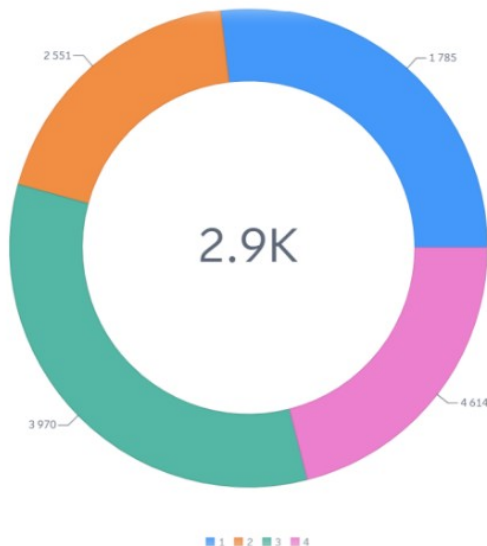
Gender	Payment Method	Season	Shipping Type
Male	Debit Card	Spring	Express
Male	PayPal	Spring	Free Shipping
Male	Credit Card	Winter	Standard

Size
M
L
M

all clusters are more than 10% of the dataset, so there is no outlier cluster, and the smallest cluster is Cluster 2 with **814** customers, and it is greater than 10% of the dataset.

Variable	Cluster 1	Cluster 2	Cluster 3
Age	Highest	Lowest	Medium
Previous Purchases	High	Low	High
Purchase Amount	Low	Low	Highest
Review Rating	Medium	Highest	Medium
Category	Clothing	Accessories	Clothing

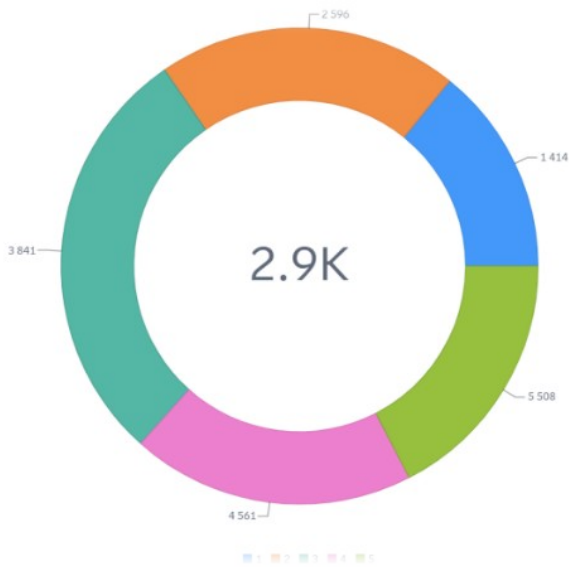
km_4 Segmentation:



Cluster	Size
Cluster 1	785
Cluster 2	551
Cluster 3	970
Cluster 4	614

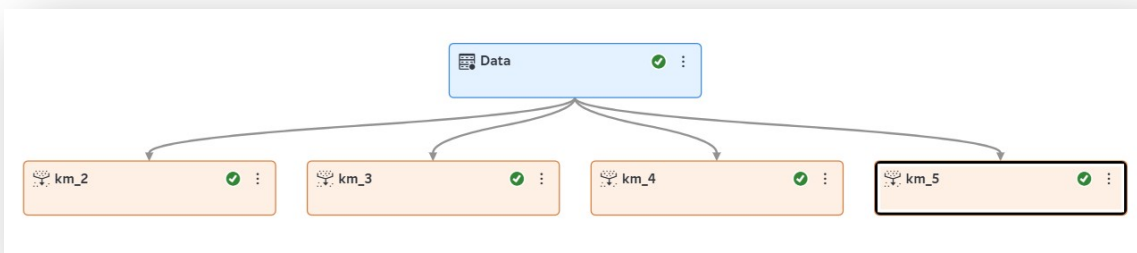
The km_4 divided the customers into four clusters with different cluster sizes

km_5 Segmentation:



Cluster	Size
Cluster 1	414
Cluster 2	596
Cluster 3	841
Cluster 4	561
Cluster 5	508

The km_5 divided the customers into five clusters with different cluster sizes.



Segmentation	Number of	Cluster Sizes	Size of Smallest Cluster	Outlier
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Label	Clusters			
km_2	2	1599, 1321	1321	No
km_3	3	1050, 814, 1056	814	No
km_4	4	785, 551, 970, 614	551	No
km_5	5	414, 596, 841, 561, 508	414	No

2. Generating Best 4 DTs

Data Partition Node:	Percentages	Cross Validation 10 folds
	Method	Stratified
Tree Node	Splitting Criterion	CHAID
		Entropy
		Gini
		Information gain ratio
	Minimum Number of Observations per Leaf	30% of smallest cluster = 244 observations
	Maximum number of branches:	2
		4

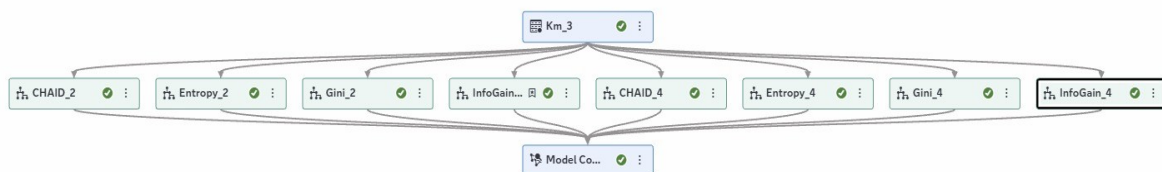


Figure 8 - Decision Tree and Model Comparison Pipeline

Decision Tree Comparison

- The decision trees were compared using accuracy and simplicity.
- Accuracy was weighted as 0.60 and simplicity was weighted as 0.40.
- The number of leaves after pruning was used to measure simplicity.

Decision Tree (CHAID_2):

Accuracy Score = 0.9466
Simplicity Score = 1.0000
Overall Score = 0.9680

Decision Tree (Entropy_2):

Accuracy Score = 0.9466
Simplicity Score = 0.5000
Overall Score = 0.7680

Decision Tree (Gini_2):

Accuracy Score = 0.9466
Simplicity Score = 1.0000
Overall Score = 0.9680

Decision Tree (InfoGain_2):

Accuracy Score = 0.9466
Simplicity Score = 0.7500
Overall Score = 0.8680

Decision Tree (CHAID_4):

Accuracy Score = 0.9466
Simplicity Score = 0.7500
Overall Score = 0.8680

Decision Tree (Entropy_4):

Accuracy Score = 0.9466
Simplicity Score = 0.0000
Overall Score = 0.5680

Decision Tree (Gini_4):

Accuracy Score = 0.9466
Simplicity Score = 0.2500
Overall Score = 0.6680

Decision Tree (InfoGain_4):

Accuracy Score = 0.9466
Simplicity Score = 0.7500
Overall Score = 0.8680

Based on the overall score comparison, the best-performing models are CHAID_2 and Gini_2, both achieving the highest overall score of 0.9680. These models had the lowest number of leaves while keeping the same accuracy score as the other models. This shows that CHAID_2 and Gini_2 gave a good balance between accuracy and simplicity

Variable	Cluster_0	Cluster_1	Cluster_2	Explanation
Previous Purchases	Medium	Low	High	This was the most important variable. Cluster 2 has the highest previous purchases, while Cluster 1 has the lowest.
Review Rating	Low	High	Medium	Cluster 1 has the highest review rating, while Cluster 0 has the lowest.
Purchase Amount (USD)	Low	Medium	High	Cluster 2 has the highest purchase amount, while Cluster 0 has the lowest.

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InfoGain_2, Entropy_2, Gini_2, and InfoGain_4 achieved the highest overall score of 0.8503. These models gave a good balance between accuracy, lift, simplicity, and stability. The most important variables were Previous Purchases, Review Rating, and Purchase Amount. These variables helped separate the customer clusters based on shopping behavior

Variable Importance

Variable Name	Training Importance	Training Relative Importance	Validation Relative Importance
Previous Purchases	395.4743	1	1
Review Rating	204.2171	0.5164	0.4497
Purchase Amount (USD)	94.1194	0.2380	0.2274

Figure 9 - Variable Importance of the Best Decision Tree

Task 3

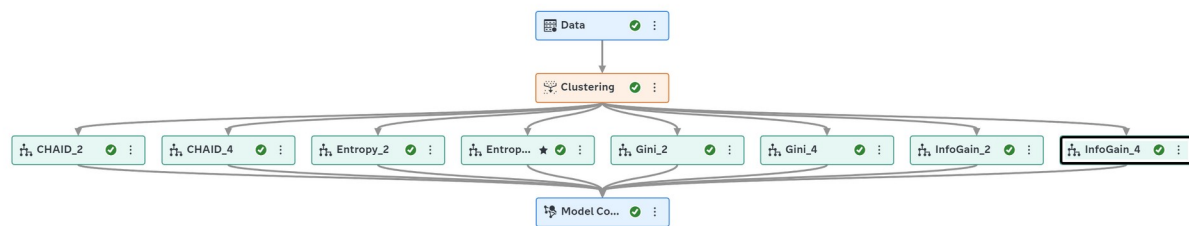


Figure 8 - Decision Tree Cluster Flow

We generated eight different decision tree models to compare performance under varying numbers of branches. This was done because the number of branches differs between the models (specifically 2 and 4 branches), and a minimum leaf size of 244 observations was used.

DT For Cluster Analysis

Decision Tree(CHAID_2):

Accuracy Score = 0.7864
Simplicity Score = 1.0
Lift Score = 0.0
Stability Score = 1
Overall Score = 0.6718

Decision Tree (CHAID_4):

Accuracy Score = 0.7864
Simplicity Score = 1.0
Lift Score = 0.0
Stability Score = 1
Overall Score = 0.6718

Decision Tree (Entropy_2):

Accuracy Score = 0.7782
Simplicity Score = 0.6667
Lift Score = 1.0
Stability Score = 1
Overall Score = 0.8503

Decision Tree (Entropy_4):

Accuracy Score = 0.7782
Simplicity Score = 0.0
Lift Score = 0.5426
Stability Score = 1
Overall Score = 0.7254

Decision Tree (Gini_2):

Accuracy Score = 0.7782
Simplicity Score = 0.6667
Lift Score = 1.0
Stability Score = 1
Overall Score = 0.8503

Decision Tree (Gini_4):

Accuracy Score = 0.7782
Simplicity Score = 0.0
Lift Score = 0.5426
Stability Score = 1
Overall Score = 0.7254

Decision Tree (InfoGain_2):

Accuracy Score = 0.7782
Simplicity Score = 0.6667
Lift Score = 1.0
Stability Score = 1
Overall Score = 0.8503

Decision Tree (InfoGain_4):

Accuracy Score = 0.7782
Simplicity Score = 0.6667
Lift Score = 1.0
Stability Score = 1
Overall Score = 0.8503

Based on the overall score comparison, the best-performing models are Decision Tree (7), Decision Tree (2), Decision Tree (4), and Decision Tree (6), all achieving the highest overall score of 0.8503, indicating a strong balance between accuracy, lift, simplicity, and stability.

Comparison with Task 2

In Task 2, all DT achieved the same accuracy of 0.9466. In Task 3, CHAID_2 and CHAID_4 achieved the highest accuracy of 0.7864. So, this shows that the DTs in Task 2 achieved higher accuracy than the DT from Task 3.

Task	Models	Accuracy Score	Result
Task 2	All DT	0.9466	High accuracy
Task 3	CHAID_2, CHAID_4	0.7864	High accuracy

Conclusion

In conclusion, the project aims to implement data mining techniques to show the customer behaviour based on Subscription status. In task 1, we generated sixteen decision trees with different splitting criteria and leaf sizes. The best decision trees are DT7 and DT11, which achieved the highest overall score of 0.841. In task 2, customer segmentation was generated using K-Means clustering with different cluster numbers. The clusters were compared based on customer behavior, purchase amount, and review rating. The KM_3 model gave balanced clusters and helped identify different shopping behavior patterns without generating outlier clusters. Finally, in task 3, We created four segmentations and focused on KM_3 to describe the clusters. The results showed that the clusters were balanced, and no outlier was found. Also, Task 2 gave higher accuracy than Task 3. eight trees were generated based on the cluster number and compared using the combination function, the best-performing models are Decision Tree (7), Decision Tree (2), Decision Tree (4), and Decision Tree (6), all achieving the highest overall score of 0.8503.